

# Fernando Manuel Sáenz

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## Profile

I am a senior software engineer and engineering leader with 17+ years of experience designing and developing mission-critical, high-availability systems. I have led the construction and optimization of complex monoliths (Ruby on Rails, Node.js) and the migration toward microservices architectures across various stacks (Ruby, Node, Python, Go, .NET), always selecting the most appropriate technological solution for each challenge. I have also managed high-performing, distributed engineering teams, defining architecture, development processes, and culture from the ground up. I am passionate about tackling large-scale technical challenges and building platforms that seamlessly integrate scalability, efficiency, and security to drive tangible business results.

## Experience

### **ENGINEERING MANAGER – LIBERTIFY, MADRID (HYBRID) • 09/2025 - PRESENT**

Libertify is an AI-powered video sharing and content generation platform that enables creators to produce and distribute high-quality multimedia content at scale.

- Lead a distributed frontend team of 7 engineers split across Madrid and Tel Aviv, fostering a strong engineering culture focused on ownership and continuous improvement.
- Define the frontend architecture and technical roadmap, establishing development standards and best practices aligned with the company's growth trajectory.
- Designed and implemented core engineering processes from scratch, including performance review frameworks with role definitions and clear progression paths, on-call rotation systems, and migration from a traditional Git workflow to trunk-based development.
- Oversee the development of the main web application in React and lead the design and implementation of a new React Native mobile application.
- Collaborate closely with product, design, and backend teams to align technical decisions with business objectives and ensure timely delivery.

### **Skills:**

- Languages: TypeScript, JavaScript
- Frameworks: React, React Native
- Practices: Engineering Management

### **INDEPENDENT CONSULTANT – MAXAM, MADRID (REMOTE) • 01/2025 - 08/2025**

MAXAM is a global leader in energetic materials and blasting solutions, operating in mining, construction, and defense industries across more than 50 countries.

- Designed and implemented a real-time data solution to feed the mathematical models used for explosion calculation and simulation, enabling more accurate and timely decision-making in the field.

- Architected a scalable Data Lake from the ground up, defining the ingestion strategy, data modeling, and ETL pipelines to consolidate structured and unstructured data sources into a unified, queryable layer.
- Designed and built end-to-end ETL processes to normalize, validate, and transform raw operational data, making it reliably available for downstream analytical and scientific processing workloads.
- Collaborated with data scientists and domain experts to understand the requirements of the mathematical models, ensuring data quality, latency, and schema consistency met the computational demands.

#### Skills:

- Languages: Python
- Cloud: Azure (Data Lake Storage, Data Factory, Azure Databricks)
- Practices: Data Lake Architecture, ETL Design, Real-Time Data Pipelines

#### **HANDS ON CTO, UNICITI, MADRID (REMOTE) – 06/2023 - 01/2025**

- Designed and implemented the company's first mobile application using React Native, focused on geolocation and real-time security alerts.
- Developed the backend in Ruby on Rails, utilizing Action Cable for live updates and SolidQueue for background tasks, delivering a smooth and consistent tracking experience.
- Built an asynchronous processing pipeline with SolidQueue to decouple the handling of new locations from request logic, reducing API latency. Implemented the Haversine algorithm combined with a custom algorithm to calculate routes and accurately determine transportation modes in real time.
- Deployed the platform on Heroku, optimizing the infrastructure to scale based on demand and ensure high availability.
- Automated app releases on Google Play and the Apple Store using Fastlane pipelines, streamlining the build and release process.
- Implemented a system of real-time notifications and security alerts based on geolocation data, improving immediate responses during critical situations.
- Collaborated closely with the product team to define key functionalities, contributing technical insights and ensuring the feasibility of requested features.
- Developed and launched the MVP in less than 6 months, demonstrating the ability to iterate quickly and deliver market value.

#### Skills:

- Languages: Ruby, Javascript, TypeScript
- Frameworks: Ruby on Rails, React Native
- Databases: PostgreSQL
- Monitoring: Sentry

- CI/CD: Github Actions, Fastlane

#### **VP OF ENGINEERING, CONEKTA, MADRID (REMOTE) – 05/2022 - 06/2024**

Conekta is a Mexican fintech specializing in online payment processing, handling millions of dollars in annual transactions and connecting merchants with multiple payment methods.

- Led a department of 150 engineers, organized into specialized teams (Data, SRE, Security, Development) and small squads, fostering autonomy and ownership within each group.
- Implemented a DevOps culture and Agile methodologies by replacing a legacy AWS CodeDeploy pipeline (which required special permissions and complicated rollbacks) with a Jenkins pipeline accessible to the entire team. This included automated validations (unit tests, security scans, and dependency checks) that reduced rollback complexity and reinforced the “You build it, you run it” mentality.
- Restructured the technical organization to reinforce end-to-end collaboration and accountability—from feature planning to implementation and production operation.
- Scaled the payment platform reliably, increasing availability from 94% to 99.99% and processing millions of dollars annually without interruptions.
- While maintaining a hands-on approach, participated in critical architecture reviews and supervised improvements in resilience and security, promoting continuous innovation.
- Managed a hybrid architecture composed of a Ruby on Rails monolith and over 80 microservices built in Go, .NET, Python, Node, and Ruby, defining quality guidelines and deployment standards for each stack.

#### **DIRECTOR OF ENGINEERING, CONEKTA, SANTIAGO DE CHILE (REMOTE) – 11/2021 - 05/2022**

- Oversaw the evolution of systems and led the implementation of large-scale new functionalities, ensuring continuous availability for clients with high transaction volumes.
- Headed the integration of AMEX cards, managing everything from technical planning and certification to final deployment, adding a critical payment method for business growth.
- Re-architected cash services with a provider-agnostic approach, facilitating the onboarding of new providers and accelerating integrations.
- Maintained an active role in code reviews and architectural decisions, promoting quality and consistency across technical solutions.

#### **PRINCIPAL ENGINEER, CONEKTA, SANTIAGO DE CHILE (REMOTE) – 02/2020 - 11/2021**

- Assumed technical leadership in refactoring a large Ruby on Rails monolith with reliability issues, transforming it into a more stable and maintainable system.
- Introduced a trunk-based development strategy, enabling collaboration across multiple teams on the same codebase and reducing merge conflicts.
- Designed and implemented a Main Event Bus using Kafka Connect to offload the primary database (MongoDB), shifting report and listing processing to asynchronous tasks and enhancing overall performance.

- Optimized the architecture and automated testing practices, increasing the reliability of the payment system and enabling iterative delivery of new features.

#### Skills:

- Languages: Ruby, Node, Javascript, Python, Go
- Frameworks: Ruby on Rails, FastAPI, Express
- Databases: MongoDB, PostgreSQL, RedShift, DynamoDB
- Cloud: AWS (ECS, RDS, Lambda, CloudWatch, KMS, SQS, API Gateway, SNS, VPC, Kinesis, DMS)
- Monitoring: New Relic, DataDog, Prometheus (Grafana)
- CI/CD: Jenkins, TeamCity

#### **TECH LEAD, DEVBORN, BUENOS AIRES (REMOTE) – 01/2018 - 02/2020**

- A. Sibly (<https://www.sibly.com/>): A mental health platform based on mobile applications, focused on real-time support and counseling for users.
  - Refactored and sanitized a monolith in Node.js with Express, initiating the transition toward a microservices architecture in AWS (primarily using Lambdas and API Gateway).
  - Designed and maintained the infrastructure—including ECS, SQS, and RDS with MariaDB—to support both the monolithic version and the new microservices, ensuring scalability and high availability across environments.
  - Developed the deployment strategy and configured multiple development environments, taking on infrastructure responsibilities (in the absence of a dedicated team) and providing continuous support to the React Native mobile app.
- B. Habit (<https://habit.com/>): A personalized nutrition service that combines genetic analysis with health and diet recommendations.
  - Maintained and evolved a Ruby on Rails monolith that not only provided services to the mobile app but also processed genetic data (DNA/proteins) within the backend to generate personalized health recommendations.
  - Improved system performance and stability by making architectural adjustments and refining CI/CD practices to handle an increasing volume of users and data.
  - Collaborated closely with product and development teams to define technical priorities and guide the implementation of new features.

#### Skills:

- Languages: Ruby, Node, Javascript
- Frameworks: Ruby on Rails, React, React Native
- Databases: PostgreSQL, MariaDB
- Cloud: AWS (ECS, RDS, Lambda, CloudWatch, KMS, SQS, API Gateway, SNS, VPC)
- Monitoring: New Relic

- CI/CD: BitRise, CircleCI

#### **SOFTWARE ARCHITECT, TRUELOGIC, BUENOS AIRES – 03/2016 - 01/2019**

A. VTS (<https://www.vts.com/>): A commercial real estate platform focused on space management and data integration, enabling property owners and brokers to make real-time, informed decisions.

- Maintained and evolved a Ruby on Rails monolith deployed on Heroku, with a React frontend.
- Integrated large volumes of data from external clients by developing tooling and processes to ingest information from various providers, ensuring high performance and reliability.
- Implemented and optimized Sidekiq for background processing, actively managing memory and load distribution to avoid bottlenecks.
- Collaborated with the integrations team to define performance strategies, adjusting database indexes, data structures, and Rails configurations to process large volumes without degrading the user experience.
- Monitored and debugged response times using logging and profiling tools, identifying bottlenecks and applying continuous improvements to the architecture and code.

#### **Skills:**

- Language: Ruby
- Frameworks: Ruby on Rails, Sinatra
- Databases: PostgreSQL, MongoDB
- Cloud: Heroku
- Monitoring: New Relic
- CI/CD: CircleCI

#### **TECH LEAD, DESPEGAR.COM, BUENOS AIRES – 09/2011 - 02/2017**

Despegar.com is the largest online travel agency in Latin America, offering flight, hotel, and tour package bookings across multiple countries in the region.

- Led the Payments team, developing and integrating payment services for Chile, Argentina, Colombia, and Ecuador while ensuring PCI compliance and the security of sensitive data.
- Implemented solutions in Ruby to connect with various payment providers, ensuring high levels of availability and reliability in transactions.
- Designed a dedicated monitoring, configuration, and logging environment (using Ruby on Rails) for the payment platform, facilitating early incident detection and real-time troubleshooting.
- Coordinated with internal teams (product, operations, and infrastructure) to align business requirements with PCI DSS compliance, maintaining a strong focus on quality and security from the outset of each project.
- Optimized the architecture of the payment systems, reducing latency and enhancing scalability as Despegar expanded its customer base.

Skills:

- Language: Ruby
- Frameworks: Ruby on Rails, Sinatra
- Databases: MySQL, MongoDB
- Monitoring: New Relic

**SOFTWARE ENGINEER, TERACODE, BUENOS AIRES – 08/2008 - 02/2011**

- Participated in various projects with responsibilities in both backend and frontend development, applying best practices and ensuring code quality throughout the product lifecycle.

B. Blue Leaf (Ruby on Rails):

- Developed a secure platform for tracking, sharing, and advising on financial records, implementing encryption and strict security policies to protect user data.
- Optimized the architecture to handle sensitive data and enhance user experience, upholding high standards of confidentiality and reliability.

B. Unithrive (Ruby on Rails):

- Implemented a solution that connected students with lenders, streamlining the process of acquiring educational loans.
- Designed user workflows and interfaces in collaboration with design and product teams to ensure an intuitive and efficient user experience.
- Collaborated closely with multidisciplinary teams to define requirements, conduct testing, and deploy applications in production environments.
- Applied Agile methodologies and version control best practices, contributing to the continuous delivery of features and patches.

Skills:

- Language: Ruby
- Frameworks: Ruby on Rails
- Databases: MySQL
- CI/CD: CircleCI
- Monitoring: New Relic

**SOFTWARE ENGINEER, MERCADOLIBRE, BUENOS AIRES – 03/2007 - 08/2008**

MercadoLibre is the leading e-commerce platform in Latin America, offering services for buying, selling, and online payments, as well as logistics and financing solutions.

- Automated tests and operational processes for various web applications and internal business workflows using Ruby as the primary language.

- Designed and implemented scripts and monitoring tools that reduced the time spent on manual validations and improved deliverable quality.
- Participated in code reviews and requirements analysis, ensuring the implementation of effective and maintainable long-term solutions.

#### Skills

- Languages: Ruby, Python
- Frameworks: Ruby on Rails
- Databases: MySQL, Oracle
- Monitoring: New Relic

#### Education

Universidad Tecnológica Nacional. Bachelor's in Information Systems Engineering – 2014

#### Certifications

Stanford-Endeavor Leadership Development Program. – 2021

Master The Power MBA, The Power MBA. – 2025